

Automated Diabetic Retinopathy Screening in Out-patient Diabetes Care – Comparison of Two Artificial Intelligence Algorithms: RetCAD and OphtAI

Automatisierte Vorsorgeuntersuchung auf diabetische Retinopathie in einer diabetologischen Ambulanz – Vergleich von 2 auf künstlicher Intelligenz basierenden Algorithmen RetCAD und OphtAI



Authors

Florian Maria Bauer^{1,2}, Annette Sauerbeck^{2,3}, Wolfgang Hitzl^{4,5,6}, Nick Piravej^{1,2}, Josef Schmidbauer^{1,2}

Affiliations

- 1 Augenheilkunde, Klinikum Nürnberg, Standort Nord, Nürnberg, Deutschland
- 2 Augenklinik, Paracelsus Medizinische Privatuniversität Nürnberg, Deutschland
- 3 Ambulantes Behandlungszentrum, Diabetologie, Klinikum Nürnberg, Standort Nord, Nürnberg, Deutschland
- 4 Forschungs- und Innovationsmanagement, Biostatistik, Paracelsus Medizinische Privatuniversität Salzburg, Österreich
- 5 Universitätsklinik für Augenheilkunde und Optometrie, Paracelsus Medizinische Privatuniversität Salzburg, Österreich
- 6 Forschungsprogramm Experimentelle Ophthalmologie und Glaukomforschung, Medizinische Privatuniversität, Salzburg, Österreich

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Georg Thieme Verlag KG, Oswald-Hesse-Straße 50, 70469 Stuttgart, Germany

Correspondence

Dr. Florian Maria Bauer, MD
Augenheilkunde, Klinikum Nürnberg, Standort Nord
Prof.-Ernst-Nathan-Str. 1, 90419 Nürnberg, Deutschland
Phone: + 49 (0) 91 13 98 25 14/76 52, Fax:
+ 49 (0) 91 13 98 76 50
florian_maria.bauer@yahoo.com

ABSTRACT

Objective The artificial intelligence (AI) can be applied to screening for diabetic retinopathy (DR) from colour fundus photographs. The prerequisite for this is that the AI used can achieve a similar performance in the real world in different study conditions. The aim of this study is therefore to test and compare the latest version of the AI-based algorithms RetCAD and OphtAI for DR screening in a diabetes outpatient clinic.

Methods In the period from August 2023 to November 2023, 150 diabetics were recruited at the outpatient diabetes center of the University Hospital. For each study participant, images were taken with the handheld retinal camera Aurora (Optomed Plc, Oulu, Finland) in Miosis. The images were examined by the ophthalmologist and by the AI-based algorithms RetCAD version 2.2.0 (Thirona Retina, Nijmegen, Netherlands) and OphtAI version 2.3.4 (Groupe Evolucare Technologies, Le Pecq, France) for the presence of DR. The severity of DR was classified using the International Clinical Diabetic Retinopathy (ICDR) scale. Patients with no retinal changes or a mild DR were advised to have an ophthalmological check-up in one year. In the presence of a moderate, severe or proliferative DR, a referral to the treating ophthalmologist was made. For this reason, the severity levels of moderate, severe and proliferative DR have been summarised under the umbrella term of referable DR.

Results No DR was detected in 123 out of 143 (86.0%) diabetics and mild DR was detected in 10 (7.3%). All patients with moderate DR 7 (5.0%), severe 2 (1.5%) and proliferative DR 1 (0.7%) were grouped together as referable DR and

represented a proportion of 7.3%. The AI-based algorithm RetCAD version 2.2.0 achieved a sensitivity of 90% and a specificity of 100% for the detection of a referable DR compared to ophthalmological image assessment. RetCAD rated 98% of the images for image analysis as sufficient or better. In contrast, the second AI-based algorithm OphtAI version 2.3.4 achieved a sensitivity of 70% and a specificity of 100% for the detection of a referable DR. The OphtAI software was able to perform image analysis on all images.

Conclusion The results for the detection of a referable DR were consistent under study conditions and in clinical use for the AI-based algorithm RetCAD. The AI-based algorithm OphtAI, on the other hand, detected fewer patients with moderate DR, which was reflected in lower sensitivity. Both algorithms correctly assigned all patients with severe and proliferative DR. The AI-based algorithms RetCAD and OphtAI tested appear to be suitable for use in a diabetes outpatient clinic and primary care setting, respectively.

ZUSAMMENFASSUNG

Hintergrund Der Einsatz von künstlicher Intelligenz (KI) bietet sich für die Früherkennung der diabetischen Retinopathie (DR) anhand von Netzhautbildern an. Voraussetzung ist, dass die eingesetzte KI, die unter Studienbedingungen erzielte Performance auch unter Real-World-Gegebenheiten erreicht. Ziel dieser Studie ist es, die auf KI-basierenden Algorithmen RetCAD und OphtAI in der aktuellsten Version für die Vorsorgeuntersuchung auf DR in einer diabetologischen Ambulanz zu testen.

Patienten und Methoden Im Zeitraum von August bis November 2023 wurden 150 Personen mit Diabetes im ambulanten Behandlungszentrum der Diabetologie am Universitätsklinikum rekrutiert. Bei jedem Studienteilnehmer wurden Fundusbilder mit der mobilen Kamera Aurora (Optomed Plc, Oulu, Finnland) in Miosis aufgenommen. Die Aufnahmen wur-

den vom Augenarzt und von den auf KI basierenden Algorithmen RetCAD Version 2.2.0 (Thirona Retina, Nijmegen, Niederlande) und OphtAI Version 2.3.4 (Groupe Evolucare Technologies, Le Pecq, Frankreich) auf das Vorliegen einer DR untersucht. Der Schweregrad der DR wurde anhand der internationalen Klassifikation eingeteilt. Personen mit Diabetes ohne DR oder einer milden DR wurde eine augenärztliche Kontrolle in einem Jahr empfohlen. Bei Vorliegen einer moderaten, schweren oder proliferativen DR erfolgte eine Überweisung an den behandelnden Augenarzt.

Ergebnisse Bei 123 von 143 (86,0%) Personen mit Diabetes wurde keine DR und bei 10 (7,3%) eine milde DR erkannt. Alle Patienten mit einer moderaten DR 7 (5,0%), einer schweren 2 (1,5%) und einer proliferativen DR 1 (0,7%) wurden als überweisungsbedürftige DR zusammengefasst und stellten einen Anteil von 7,3% dar. Der auf KI basierende Algorithmus RetCAD Version 2.2.0 erreichte eine Sensitivität von 90% und eine Spezifität von 100% für die Erkennung einer überweisungsbedürftigen DR im Vergleich zur augenärztlichen Bildbeurteilung. RetCAD bewertete 98% der Bilder für die Bildanalyse als ausreichend oder besser. Der zweite KI-basierte Algorithmus OphtAI Version 2.3.4 erzielte eine Sensitivität von 70% und eine Spezifität von 100% für die Detektion einer überweisungsbedürftigen DR. Eine Bildanalyse konnte bei allen Aufnahmen durchgeführt werden.

Schlussfolgerung RetCAD erreichte im klinischen Einsatz für das Erkennen einer überweisungsbedürftigen DR die unter Studienbedingungen angegebene Performance. Der zweite Algorithmus OphtAI erkannte weniger Patienten mit moderater DR, was sich in einer geringeren Sensitivität widerspiegelte. Von beiden Algorithmen wurden alle Patienten mit schwerer und proliferativer DR richtig zugeordnet. Die getesteten KI-basierten Algorithmen RetCAD und OphtAI scheinen für den Einsatz in einer Diabetesambulanz beziehungsweise in der Primärversorgung geeignet.

Introduction

The prevalence of diabetes mellitus is about 8.9% of the adult population in Germany [1]. Diabetic retinal changes can be detected in as many as one in five people with diabetes. Diabetic retinopathy (DR) is the main cause of avoidable visual impairment and blindness in people of working age. If the retinal disorder is detected early, the prognosis is very good because of the potential for treatment with laser coagulation and intravitreal drug administration [2].

For this reason, screening for DR is of critical importance, as patients usually only notice visual impairment at an advanced stage of the disease. In Germany, retinal examinations are recommended every one to two years depending on the risk constellation.

In 2020, in North Rhine-Westphalia only 66.7% of people with diabetes had their two-year ophthalmological screening as recommended by the Disease Management Program (DMP) [1].

From 2002 to 2017, the number of patients with DR in Germany increased by 15%, while the ophthalmological care capacity increased by only 1% during the same period [3].

The use of artificial intelligence lends itself to screening, as retinal imaging combined with an AI-based image analysis algorithm allows for quick and resource-efficient examination [4]. This could enable the screening to be carried out during the doctor's appointment with the diabetologist or family doctor. This additionally increases the willingness of patients to participate in eye screening [5,6]. If moderate, severe, or proliferative DR were detected during automated screening, patients could be referred to their treating ophthalmologist for further assessment.

In Germany, AI-based screening for diabetic retinopathy is viewed positively and is considered 'fundamentally suitable for future use' according to the National Care Guideline [1]. However, fundus photography, which is required for AI-based screening for diabetic retinopathy, is currently not covered by statutory health insurance and is considered an out-of-pocket service (IGeL).

In the United States of America, the first automated screening system for DR was approved back in 2018 [7]. The payment for this benefit is made through the health insurance provider and has been covered in the USA since 2021 [8].

Many studies prove that AI-supported image assessment is equivalent to ophthalmological assessment [4]. However, there is evidence in the literature that the algorithms perform worse under real-world conditions compared to results obtained under study conditions [9, 10].

Due to the continuous optimisation of the algorithms, it is expected that performance under real-world conditions will continue to improve.

The aim of this study is therefore to test and compare the latest versions of the two commercially available AI-based algorithms RetCAD and OphtAI under real-world conditions.

Materials and Methods

study was carried out in accordance with the Helsinki Declaration and after a positive vote of the Institutional Ethics Committee of the University Hospital. From August to November 2023, 150 patients with known type 1 or type 2 diabetes mellitus were recruited in the scope of their diabetic follow-up examination. The only inclusion criteria were known diabetes mellitus and legal age of majority. Participants were informed about the study and signed a written consent form. First, the diabetic examination was carried out in the outpatient treatment centre and then the screening examination for diabetic retinopathy was carried out on site.

For this purpose, the hand-held camera Aurora (Optomed Plc, Oulu, Finland) was used. The device has a CE mark and is the only mobile camera that is approved in combination with the algorithm of AEYE-DS (AEYE Health Inc, Tel Aviv, Israel) as an automated screening system for DR in the USA. The camera is suitable for shooting done in miosis and has a field of view of 50 degrees.

A total of four images per patient were taken in a darkened room in the diabetes outpatient clinic. One photo with the optic disc and one with the macula in the centre of the image. If the image quality was poor, the images were taken again. If it was still not possible to obtain retinal photos in miosis in sufficient image quality, the patients were offered an examination in mydriasis.

All images were stored in the web-based data management software Harmony (Topcon Medical Systems, Oakland, USA).

In a first step, all retinal photos were assessed by the ophthalmologist. The DR was classified according to the ICDR classification [11]. Ophthalmological monitoring in one year was recommended for patients with no retinal changes or mild DR.

If moderate, severe or proliferative DR was observed, the patient was referred to the treating ophthalmologist. For this reason, the severity levels moderate, severe and proliferative DR have been grouped together under the umbrella of referable DR.

The patient was classified according to the eye with more severe DR.

In a second step, the latest version of the AI-based software RetCAD v.2.2.0 (Thirona Retina, Nijmegen, Netherlands) was used [12, 13]. It is a class IIa medical device that has been certified in the EU since 2022. Certification refers to the software and is independent of which camera was used. RetCAD v.2.2.0 is an AI-based

program based on deep learning that uses multiple algorithms to assess image quality and classify DR. Image quality from a value of less than 25 is considered insufficient for image analysis.

The DR was divided into five severity levels by the AI-based algorithm analogous to the ICDR classification: If the score is less than 1, there is no DR. A score of 1 to less than 2 indicates the presence of mild DR. A score of 2 to less than 3 is moderate DR. Severe non-proliferative DR is present at a value of 3 to less than 4 and proliferative DR is present at 4 and above.

A report for the patient was prepared if at least one fundus photo of sufficient image quality was available for each eye [14, 15] (► Fig. 1).

In addition, retinal areas identified as abnormal by the AI-based software on fundus photographs were highlighted in colour and displayed as a heatmap.

In a third step, the AI-based algorithm OphtAI version 2.3.4 (Groupe Evolucare Technologies, Le Pecq, France) was applied. It is also a class IIa medical device.

OphtAI version 2.3.4 is an AI-based program that is based on deep learning and also uses several algorithms to assess the image quality and the classification of DR. Image quality is divided into two levels: good and poor. DR was classified into five severity levels based on the ICDR classification.

Furthermore, all conspicuous retinal areas were colour-marked by the AI-based algorithm and displayed in the form of a heatmap. A report was created for each fundus photo (► Fig. 2).

At the end of the image analysis, every study participant with referable DR or notable incidental findings received a written notification. The treating ophthalmologist was also informed.

The following statistical methods were used: The data were checked for consistency, sensitivity, specificity, negative and positive predictive value and overall accuracies were calculated with associated 95% confidence intervals. The data were analysed using STATISTICA 13 (Hill, T. & Lewicki P. Statistics: Methods and Applications. StatSoft, Tulsa, OK).

Results

In 143 of 150 study participants, the image data collected could be evaluated. Seven patients were excluded for the following reasons: In five individuals with diabetes, it was not possible to take shots in miosis, and they refused to be tested in mydriasis. Two patients had fewer than four shots taken.

The study population of the diabetology clinic at the outpatient treatment centre at the University Hospital was composed as follows: The average age of patients was 56 years (range 21 to 98 years). The proportion of women was 42%, of men 58%. Of these patients, 29% had type 1 diabetes mellitus and 71% had type 2 diabetes mellitus. The mean HbA1c was 7.4% (range 4.8% to 12.3%) (► Table 1).

No DR was detected in 123 of 143 (86.0%) diabetic subjects, and mild DR was detected in 10 (7.3%). All patients with moderate DR (7; 5.0%), severe DR (2; 1.5%), and proliferative DR (1; 0.7%) were grouped as referable DRerral, representing a total proportion of 7.3%. The results of the AI-based algorithms for screening for referable DR were presented clearly as contingency tables (► Tables 2 and 3).

Recommendation for the patient

Result of the RetCAD analysis
More than mild DR detected.

Recommendation
Refer the patient for further evaluation.



Image quality of photos: **moderate**

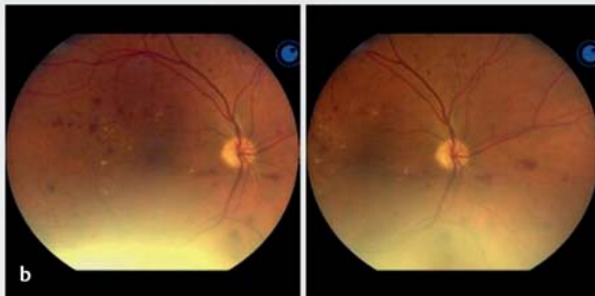
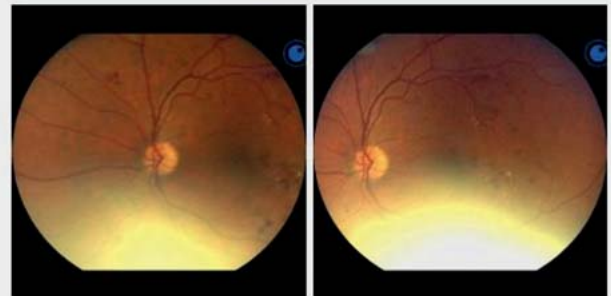


Image quality of photos: **moderate**



► **Fig. 1** a Example of a RetCAD analysis report at patient level. b Fundus images for ► **Fig. 1** a, with assessment of the image quality.

DIABETIC RETINOPATHY GRADING

RE Image 1



Pathology

DR Grading
Severe

0% No DR

0% Mild

0% Moderate

96% Severe

4% Proliferative

► **Fig. 2** Example of a report of the OphtAI analysis for the right eye, image with the optic nerve in the centre.

► **Table 1** Characteristics of the study population.

Characteristics of the study population	Arithmetic tool
Age in years	56 (21–98) ¹
Female	58 (42%) ²
Male	80 (58%) ²
Type 1 diabetes mellitus	40 (29%) ²
Type 2 diabetes mellitus	98 (71%) ²
HbA1c in %	7.4 (4.8–12.3) ¹
¹ min-max, ² frequency	

The artificial intelligence-based algorithm RetCAD version 2.2.0 achieved a sensitivity of 86% and a specificity of 100% for detecting moderate DR compared to ophthalmological image evaluation. All patients with both severe and proliferative DR were detected and correctly assigned by AI-based image analysis. This results in a sensitivity of 90% and specificity of 100% for referable DR (► **Table 4**). RetCAD rated the image quality as sufficient or better for analysis in 98% of the cases.

The AI-based algorithm OphtAI version 2.3.4 achieved a sensitivity of 57% and a specificity of 100% for detection of moderate DR compared to ophthalmological image evaluation. For all patients with severe and proliferative DR, there was also complete agreement between ophthalmological and AI-based image analysis. Thus, the AI-based algorithm OphtAI version 2.3.4 achieves a sensitivity of 70% and specificity of 100% for the detection of referable DR (► **Table 4**).

OphtAI version 2.3.4 rated the image quality of all images as sufficient for analysis or better.

Discussion

Automated screening for DR requires a camera and an AI-based algorithm.

Camera models that required drug pupillary dilation in a high proportion of patients have been in common use to date [16].

The AI-based algorithms are equivalent to ophthalmological image assessment under study conditions. However, there is evidence in the literature that performance may vary under real-world conditions [9, 10].

For this study, the latest versions of the AI-based algorithms RetCAD and OphtAI were tested for the first time.

Retinal imaging in miosis was possible in 95% of patients with the handheld Aurora retinal camera employed in this study. The two AI-based algorithms rated the image quality of these images as sufficient in 98% and 100% of cases.

Under real-world conditions, the AI-based algorithm RetCAD achieved a sensitivity of 90% and a specificity of 100% for the detection of referable DR.

The AI-based OphtAI software underestimated the severity in 3 of 7 people with diabetes with moderate DR. This was reflected in

► **Table 2** Contingency table of the AI-based algorithm RetCAD for screening for referable DR in 143 people with diabetes.

		Ophthalmologist Referable DR	
		Yes	No
RetCAD Referable DR	Yes	9 True positive (RP)	0 False positive (FP)
	No	1 False negative (FN)	133 True negative (RN)
		Sensitivity 9/9 + 1 = 0.9 (RP/RP+FN)	Specificity 133/133 + 0 = 1.0 (RN/RN+FP)

► **Table 3** Contingency table to examine the AI-based algorithm OphtAI for screening for referable DR in 143 people with diabetes.

		Ophthalmologist Referable DR	
		Yes	No
OphtAI Referable DR	Yes	7 True positive (RP)	0 False positive (FP)
	No	3 False negative (FN)	133 True negative (RN)
		Sensitivity 7/7 + 3 = 0.7 (RP/RP+FN)	Specificity 133/133 + 0 = 1.0 (RN/RN+FP)

a lower sensitivity of 70% for the detection of referable DR compared to ophthalmological image evaluation.

Both AI-based algorithms detected and correctly assigned all patients with severe and proliferative DR. Furthermore, in no individual with diabetes was the severity of moderate, severe, or proliferative DR overestimated.

The only study participants to be referred to a community-based ophthalmologist were those in whom the automated screening system detected referable DR, or those who could not be screened with the automated system. Thus in total, only 16 out of 150 individuals with diabetes were referred to a community-based ophthalmologist. This results in an 89% reduction in ophthalmological screening.

The hand-held camera Aurora showed a similar rate of retinal images in miosis in two studies [17, 18]. The prerequisite is that the images are taken in a darkened room.

One study compared the performance of 21 artificial intelligence (AI) algorithms for DR screening in datasets taken by hand-

► **Table 4** Diagnostic accuracy of the AI-based image analysis of RetCAD and OphtAI compared to ophthalmological image evaluation.

Classification of diabetic retinopathy (DR)	RetCAD version 2.2.0	OphtAI 2.3.4
No + mild DR	Sensitivity 100%	Sensitivity 100%
	Specificity 90%	Specificity 70%
Moderate DR	Sensitivity 86%	Sensitivity 57%
	Specificity 100%	Specificity 100%
Severe DR	Sensitivity 100%	Sensitivity 100%
	Specificity 100%	Specificity 100%
Proliferative DR	Sensitivity 100%	Sensitivity 100%
	Specificity 100%	Specificity 100%
Referable DR	Sensitivity 90%	Sensitivity 70%
	Specificity 100%	Specificity 100%

held Aurora fundus camera in a real-world setting [9]. The performance of the AI-based algorithms varied considerably: Sensitivity ranged from 13% to 97% and specificity from 20% to 100% for detection of referable DR. Only 5 of the AI providers tested achieved a good result. The identity of the manufacturers was masked and the software version of the AI-based algorithms used was not mentioned.

In a recent scientific paper, the predecessor version of the AI-based algorithm RetCAD version 2.1.1 already achieved a sensitivity of 89% and specificity of 99% for the detection of referable DR under real-world conditions [19].

However, a stationary retinal camera was used instead of the Aurora mobile camera.

Results under study conditions are available for the AI-based algorithm OphtAI. According to the manufacturer, the sensitivity for detecting diabetic retinopathy was 99%, and the specificity was 89% [20]. This performance could not be fully achieved in the current study under real-world conditions for the detection of referable DR.

With the first automated screening system approved in the United States, consisting of a Topcon pedestal device and the AI-based IDxDR, an automated screening procedure was carried out in clinical practice. Of these screenings 72% were suitable for diagnosis (14 553 out of 20 160 patients) [21]. By comparison, 95% of diabetic patients could be screened for DR using the AI-based algorithms and the Aurora mobile camera presented here, but the different size of the study populations must be taken into account in this deviation (143 out of 150 patients).

A limitation of this study is the small study population of 150 subjects. However, the study population with a share of 14.6% of persons with diabetes with diabetic retinal changes reflects the situation in Germany. The prevalence of DR was between 10% and 20% [1].

The strength of this prospective study is that people with diabetes were screened in a diabetes clinic without preselection. A

setting was thus chosen which corresponds to a future clinical application.

Further advances in camera technology are expected, making it possible to shoot in miosis in daylight. Any update of an AI-based algorithm can lead to a change in performance. For this reason, each new version must be evaluated before clinical use.

Conclusion

With the AI-based algorithms presented here, the number of ophthalmological screening examinations for DR could be reduced by 89%, but this result needs to be substantiated with higher case numbers. This good result is due to the Aurora camera, as almost all images were taken in miosis and with sufficient image quality. Also, the algorithms RetCAD and OphtAI never overestimated the severity of referable DR. All patients with severe and proliferative DR were correctly identified and assigned according to ophthalmological assessment. RetCAD and OphtAI only miscalculated in a small number of cases in detecting moderate DR.

Overall, the AI-based algorithm RetCAD demonstrated a performance under real-world conditions consistent with the results under study conditions.

The clinical use of AI-based screening for DR presented here would allow the increasing number of diabetic patients to be managed and at the same time more capacity could be created for patients with referable DR.

SUMMARY BOX

Already known:

- Diabetic retinopathy is a common complication of diabetes mellitus, so screening for detection and early treatment of DR is critical.
- The growing number of people with diabetes is placing an increasing burden on the healthcare system.
- The use of artificial intelligence is well-suited for screening, as retinal imaging combined with an AI-based image analysis algorithm allows for screening examinations to be carried out quickly and resource-consciously.
- Many studies prove that AI-supported image assessment is equivalent to ophthalmological assessment. However, there is evidence in the literature that the algorithms perform worse under real-world conditions compared to results obtained under study conditions.

Newly described:

- For this study, the latest versions of two commercially available AI-based algorithms, RetCAD and OphtAI, were tested for the early detection of diabetic retinopathy.
- The hand-held Aurora retinal camera is recommended, as almost all images could be taken in miosis and evaluated with AI support.
- RetCAD achieved a sensitivity of 90% and specificity of 100% for the detection of referable DR under real-world conditions.

- OphtAI achieved a lower sensitivity of 70% and a specificity of 100% for the detection of referable DR. However, all patients with severe and proliferative DR were correctly assigned.
- With the AI-based algorithms presented here, the number of ophthalmological screening examinations for DR could be reduced by 89%. This would help address the growing number of people with diabetes, while also increasing the capacity for patients with referable DR.

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Conflict of Interest

The authors declare that they have no conflict of interest.

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